

MANDI ERP - GRAIN MARKET MANAGEMENT SYSTEM

Complete User Manual, System Mechanics & Client Setup Guide

1. System Overview & Mandi Business Logic

The Grain Market Management System (Mandi ERP) is designed specifically for Commission Agents (Arhtai Shop / **■■■■ ■■■■■ ■■■■■**) to automate crop receiving, crop sales, multi-party ledger accounts, and custom commission/palledari deductions without manual errors.

Step	Mandi Phase	Description & Operations
Step 1	Party Accounts (■■■■■)	Setup accounts for Farmers/Sellers (Zamindar) and Buyers (Mills/Traders). Supports Opening Balances (Receivable / Payable).
Step 2	Fasal Receiving (■■■■)	Log incoming crop stock brought by Farmer (Crop type, Bags, Gross Wt, Deduction, Net Wt, Manns). No price fixed yet.
Step 3	Sales & Settlement (■■■■■ ■■■■■)	Sell crop to Buyer @ market rate. System automatically calculates Buyer Bill (+) and Farmer Net Payable (-) after deducting 3 Palledar charges (Mazdoori, Brokery, Shop).
Step 4	Ledger & Cash (■■■■■■■ ■■■■■)	Double-entry running khata logs receipts from Buyer (Vasooli) and payments to Farmer (Diye) with printable dual vouchers.

2. Financial Formulas & Calculations

Mandi ERP automates dual-entry calculation for both Buyer (Purchaser) and Farmer (Crop Seller):

BUYER TOTAL BILL FORMULA (+)	FARMER NET PAYABLE FORMULA (-)
Gross Sale = Net Weight (KG) × Rate/KG Buyer Comm = Gross Sale × Buyer Comm Rate (%) ■ BUYER TOTAL BILL = Gross Sale + Buyer Comm (Debited to Buyer Ledger: LENA HAI)	Gross Sale = Net Weight (KG) × Rate/KG (-) Farmer Comm = Gross Sale × Farmer Comm Rate (%) (-) 1. Mazdoori / Palledari = Per Bag / Rate (-) 2. Brokery / Dalali = Per Bag / Rate (-) 3. Shop / Tulai Charges = Per Bag / Rate ■ FARMER NET PAYABLE = Gross Sale - Deductions (Credited to Farmer Ledger: DENA HAI)

Practical Numerical Example:

- **Wheat Lot:** 100 Bags (4,000 KG / 100 Manns) @ Rs. 4,000/Mann (Rs. 100/KG)
- **Gross Crop Sale:** 4,000 KG × Rs. 100 = **Rs. 400,000**
- **Buyer Bill:** Rs. 400,000 + Rs. 4,000 (1% Buyer Comm) = **Rs. 404,000** (Buyer Owes You)

• **Farmer Settlement:** Rs. 400,000 - Rs. 8,000 (2% Comm) - Rs. 2,000 (Mazdoori @ Rs.20/bag) - Rs. 1,000 (Brokery) - Rs. 1,000 (Shop) = **Rs. 388,000** (You Owe Farmer)

• **Your Shop Earnings:** Rs. 4,000 (Buyer Comm) + Rs. 8,000 (Farmer Comm) + Rs. 4,000 (Deductions) = **Rs. 16,000 Net Profit**

3. Client Laptop Installation & 1-Click Desktop Setup

Follow these simple steps to set up and run Mandi ERP on your client's Windows laptop without creating a new codebase:

Step	Action Item	Instructions
1	Copy Folder	Copy the entire project folder (Grain Market Management System) to the client's laptop (e.g. D:\Grain Market Management System or C:\Mandi_ERP).
2	Install Python	Download & install Python 3.10+ from python.org. CRITICAL: Check the box 'Add python.exe to PATH' during installation.
3	One-Time Setup	Double-click on Setup_Requirements.bat inside the folder. It will install all required Python packages automatically.
4	Desktop Shortcut	Double-click on Create_Desktop_Shortcut.bat . It will instantly generate a ' Mandi ERP ' shortcut on the Windows Desktop.
5	Daily Usage	Client double-clicks on the ' Mandi ERP Desktop Shortcut ' (or Start_Mandi_ERP.bat). The Web App will auto-open in Google Chrome at http://localhost:8000 .

4. Manual Desktop Shortcut Creation (Alternative)

If you prefer to create the Windows Desktop shortcut manually:

1. Open the project folder in File Explorer.
2. Right-click on **Start_Mandi_ERP.bat**.
3. Select **Send to → Desktop (create shortcut)**.
4. Go to Desktop, right-click the shortcut, select **Rename**, and type **Mandi ERP**.

5. Features Summary for Client Presentation

- **Multi-Tenancy & Shop Profiles:** Supports multiple shops/users with separate login, shop headers, and city credentials.
- **Double-Entry Ledger:** Real-time running khata tracking exact debit, credit, opening balance, and live party status (Lena Hai / Dena Hai).
- **Dual Voucher Receipts:** 1-Click printable / shareable vouchers for both Kisan Settlement Parchi and Buyer Invoice.
- **AI Machine Learning Engine:** Built-in ML price prediction (7-day ahead crop rate forecasting) and Farmer Credit Risk scoring.
- **Offline Local Storage:** 100% offline database operation - no internet required for daily shop operations.